TerraTransfer

PRESSURE SENSOR · TYP

Ceramic pressure sensor.

MEDIA-RESISTANT · LONG-TERM STABLE · 0–1 BAR

DESCRIPTION

The **Ceramic pressure sensor Type 0420** is a rugged submersible probe with a ceramic diaphragm for the measurement of **water levels and gauges** in demanding environments. The ceramic diaphragm is particularly resistant to **aggressive media** and offers excellent **long-term stability** — even over many years of continuous service.

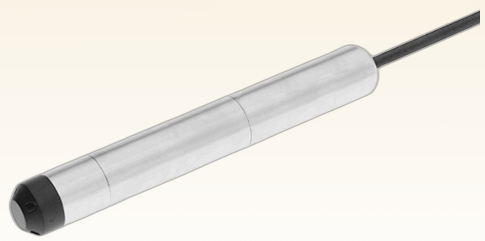
The stainless steel housing protects the electronics, the ceramic diaphragm handles the actual pressure measurement. This combination makes the sensor ideal for **leachate, industrial wastewater, treatment plants, stormwater overflow basins** and generally for waters with chemically loaded content, where conventional stainless diaphragms would degrade over time.

Communication via **SDI-12 V1.3** and in parallel via **Bluetooth 5** for on-site configuration — no app required, directly from the browser using BLX.JS. Based on the Open-SDI12-Blue platform (nRF52, < 10 µA sleep current). **IP68** for permanent immersion. Standard cable length 10 m.

APPLICATIONS

- Leachate monitoring at landfills and legacy contaminated sites
- Stormwater overflow basins and combined sewer overflows
- Wastewater treatment plants and sewer infrastructure
- Waters with aggressive content (industry, agriculture)
- Groundwater and gauging measurement with unfavorable water chemistry
- Official monitoring stations with long-term monitoring mandate
- Shallow gauges up to ≈ 10 mWC, water-level supervision

TYPE 0420 · CERAMIC



KEY FEATURES

- **Ceramic diaphragm** · media-resistant
- Range **0 – 1 bar** (≈ 10 mWC)
- Resistant to **aggressive media**
- Excellent **long-term stability**
- Housing in **stainless steel** · pressure + temperature
- Gauge pressure with capillary · **10 m cable** standard
- **SDI-12 V1.3** + **Bluetooth 5** · IP68
- **Ultra-low-power** · < 10 µA sleep current

Technical data & interfaces

CERAMIC PRESSURE SENSOR · TYPE 0420

TECHNICAL SPECIFICATION

Measuring principle	Capacitive · ceramic diaphragm · media-resistant
Measuring range	0 – 1 bar (≈ 10 mWC)
Pressure diaphragm	Ceramic — resistant to aggressive media
Housing	Stainless steel
Temperature measurement	Integrated · pressure + temperature at the same measuring point
Long-term stability	Excellent · low drift even under continuous load
Pressure type	Gauge pressure with capillary (integrated in cable)
Cable length	Standard 10 m · longer on request
Interface	SDI-12 V1.3 (3-wire: VCC · signal · GND) + Bluetooth 5 (BLE) · in parallel
Supply	2.8 V – 14 V · bus capacity up to 62 sensors · cable length > 500 m
Sleep current	< 10 µA (ultra-low-power) · BLE link ~ 20–30 µA
Platform	Open-SDI12-Blue · nRF52 SoC · ARM Cortex-M4F @ 64 MHz
Configuration	BLX.JS browser-based (Chrome, Edge, Opera, Android) · no app required · signed firmware updates
Protection class	IP68 · permanent immersion
Operating conditions	Aggressive-media resistant · suitable for leachate, wastewater, industrial waters
Compliance	CE (RED 2014/53/EU, EMC 2014/30/EU) · RoHS-compliant

Connection to Aquatos

Directly compatible with Aquatos Web LTX and mini — readings are automatically transferred to the Sensor Manager web portal, including maps, time series and alarms.

BLX.JS Dashboard

Browser-based configuration via Web Bluetooth. SDI-12 commands over BLE (prefix „z") — in parallel to the bus wiring. Chrome, Edge, Opera, Android.

Accessories

Sensor cages, clamping and mounting adapters, replacement capillaries, extended cable sets, calibration service. Installation by TerraTransfer or partners.

TerraTransfer

MADE IN GERMANY

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